Rohan Malhotra

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NYU Math and Computer Science double major (accelerated, May 2027) building quantitative, AI, and robotics systems: LLM tools and Spot robot vision at IBM, prediction-market risk research at Kalshi, and imaging tests for an ISS-deployed CubeSat.

EDUCATION

New York University, Courant Institute of Mathematical Sciences — New York, NY
B.A. in Mathematics and Computer Science (Double Major) — *Expected May 2027*

- **GPA:** 3.7/4.0 | Accelerated three-year degree | **Graduate course:** Mathematical Techniques for CS Applications

EXPERIENCE

IBM — Dallas, TX
Applications Developer Intern, IBM Consulting — *May 2026 – Aug 2026*

- Fine-tuned Qwen-Coder-32B on Oracle EPM assets, raising EPM build-generation accuracy from **36.7% to 95.0%**.
- Built an on-prem EPM assistant that turns plain-language requests into reviewable JSON using RAG and MCP tools.
- Built 13-day rolling cash forecasts, SOFR loan and interest-accrual models, and close reports in Oracle EPM.

Robotics Engineer Intern — *Jun 2026 – Aug 2026*

- Trained YOLO11 models on 898 hand-labeled images to **99.5% mAP@50** with 99.4% precision and 99.4% recall.
- Connected real-time inference to the Boston Dynamics Spot SDK via OpenCV, gRPC, and lock-free inference queues.
- Presented live retrieval demos at IBM DevCon; IBM sales and client-success teams adopted the demo.

Kalshi — New York, NY
Quantitative Developer — *Jan 2026 – May 2026*

- Built a C++ hedging engine that converts contract prices (cents, probability, bps) into real-time exposure.
- Co-authored research modeling job-loss risk as a macro-sensitive Poisson hazard with jump-diffusion income.
- Built a Next.js/Python engine that recommends hedge size and compares cost vs. tail risk using FRED and BLS data.

Hume Center for National Security and Technology, Virginia Tech — Blacksburg, VA
Aerospace Researcher — *Nov 2024 – May 2025*

- Programmed imaging and signal-processing tests in C for ContentCube, a CubeSat deployed from the ISS in 2025.
- Engineered a test environment simulating concurrent imaging workloads to resolve race conditions and memory bugs.

Aress Software — New York, NY
Machine Learning Intern — *Jun 2025 – Aug 2025*

- Trained and deployed PyTorch models on **500K+ multimodal records** across Azure and AWS.
- Built Power BI dashboards surfacing 20+ real-time KPIs for global operations teams; managed the JIRA workflow.

RESEARCH

Reddit Data in Quantitative Financial Models (published) — Jan 2025 – May 2025
Co-author, Virginia Tech (VTechWorks) — *hdl.handle.net/10919/124730*

An Economic Approach to Optimize Capital Allocation (published) — Aug 2024 – Dec 2024
Author, Virginia Tech (VTechWorks) — *hdl.handle.net/10919/124871*

PROJECTS

Refrax | refrax.io | *C++, Next.js, TypeScript, PostgreSQL* — Jun 2025 – Feb 2026

- Implemented C++ risk engines (Greeks, P&L simulation, portfolio exposure) optimized for parallel execution.

EPM Wizard | *FastAPI, React, Groovy, Playwright* — 2026

- Built a tool that turns plain-English requests into Oracle EPM forms and rules, with preview and approval steps.

Casen | getcasen.com | *JavaScript, OpenAI, Supabase, Voice AI* — 2026

- Built a case-interview prep platform with frameworks, drills, M&A news, and voice-based AI mock interviews.

NightShift | *FastAPI, Playwright, Next.js, Stripe* — 2026

- Built a platform that finds jobs, fills applications via browser automation, and verifies confirmation emails.

SKILLS

Languages: Python, C++, C, Java, SQL, TypeScript, Rust, R, MATLAB, Groovy
AI & Data: PyTorch, YOLO11, OpenCV, XGBoost, scikit-learn, LoRA, RAG, MCP, NumPy, Pandas, SciPy
Systems: Linux, Docker, Kubernetes, AWS, Azure, PostgreSQL, gRPC, WebSockets, React, Next.js, FastAPI
Finance Tools: Oracle EPM Cloud, Oracle Integration Cloud, EPM Automate, Power BI, Excel (macros)